Private Catalog Offers Refactoring Detailed Design

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Context

Feature: IPD redesign + Catalog filters

Team: Avengers

Design Story: **OBR-16351 - Getting issue details...** STATUS

Internal GetOffers v3 Detailed Design

API Git Repo: **https://github.intuit.com/WEBS/webs-pri-catalog-offers.git**

Model Git Repo: **https://github.intuit.com/WEBS/webs-pri-catalog-model-library**

Change Control

Item	Date Updated	Notes	Requestor
Created	May 17, 2018		Mariscal, Carlos
Updated	June 22, 2018	Conducted Detail Design Review	
Approved	June 22, 2018	Detail design got Approved	
Code	June 19, 2018	Development Started (API Migration to Spring Boot and Base Model Library)	
Code Complete	July 14, 2018	Code Complete	
Change requested			

Issues With Current Implementation of Private Catalog Offers

Issue No.	Issue Description	Notes

1	Hard to maintain code. Hard to enhance code.	All functionality is defined in 1 Service Class. 1,381 lines of code.
2	Too many database calls. For loop has a database call.	websitem.offer_transition is called 1,236 times taking 53,986ms for salesforce type query when calling INT-DEV. See profiler results.
3	Slow response time.	worst time in production excluding outliers(any result > 60seconds): 43485ms for salesforce type query. See splunk results table.
4	20 Minutes to run unit tests.	
5	For each offer returned in original query, calls database to find addons	for salesforce type query when calling INT-DEV. See profiler results. 14932ms = total of 26 invocations 9850ms = total of 151 invocations 4889 = total of 111 invocations 4248 = total of 94 invocations 4186 = total of 80 invocations

Visual VM Profiler Results for Salesforce Type Query

webs-pri-c...ofiler.nps

Splunk Queries for Performance

index=* success source="*webs-pri-catalog-offers-api-v3.log" kpiTotalMS < 60000 | stats max(kpiTotalMS)

index=* success source="*webs-pri-catalog-offers-api-v3.log" kpiTotalMS < 60000 | stats avg(kpiTotalMS)

index=* success source="*webs-pri-catalog-offers-api-v3.log" kpiTotalMS < 60000 | stats perc95(kpiTotalMS)

index=* success source="*webs-pri-catalog-offers-api-v3.log" kpiTotalMS < 60000 | stats perc50(kpiTotalMS)

Splunk Date - 30 days	Worst Time	Average Time	Perc95	Perc50	Query Params for Worst Time
14 June 2018	43485	259.64	470	220	offset=1000, limit=500, customer-segment=null, extended-customer-segment=null, sales-channel=null, extended-sales-channel=null, region=null, country=null, currency=null, charge-frequency=null, type=null, level=null, query-action=null, offer-ids=[], product-code=null, owned-base-offer=null, owned-addon-offers=[], list-price-offer-only=false, changeType=[], include-inactive-offers=true. intuit_locale=en_US

Analysis of OfferServiceImpl Class

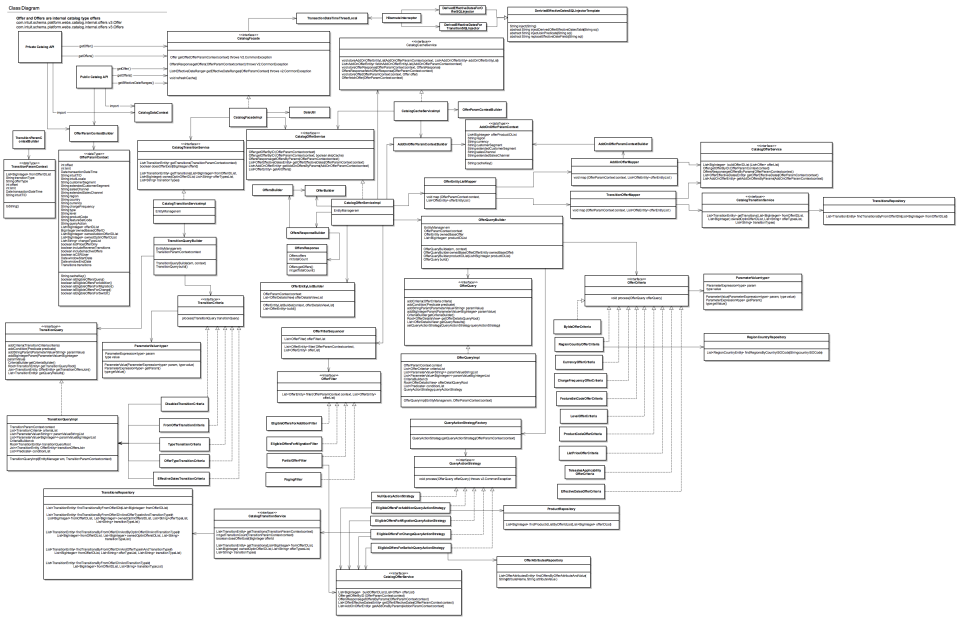
1. Call to view took 6.1 seconds on local machine to DEV-INT database for salesforce type query (This is OK for now and not where improvements are to be gained. View is required to be able to support filtering by 19 different query params)
2. Transitions were called 1,236 times and took 53.9 seconds on local machine to DEV-INT database (This was improved in Hackathon see commits. Changed algorithm to make single database call.)
3. Rest of time was calling database for looking up AddOns (Criteria can be cached to get addons list instead of calling database each time)

Proposed New Algorithm for Get Offers by Query Param

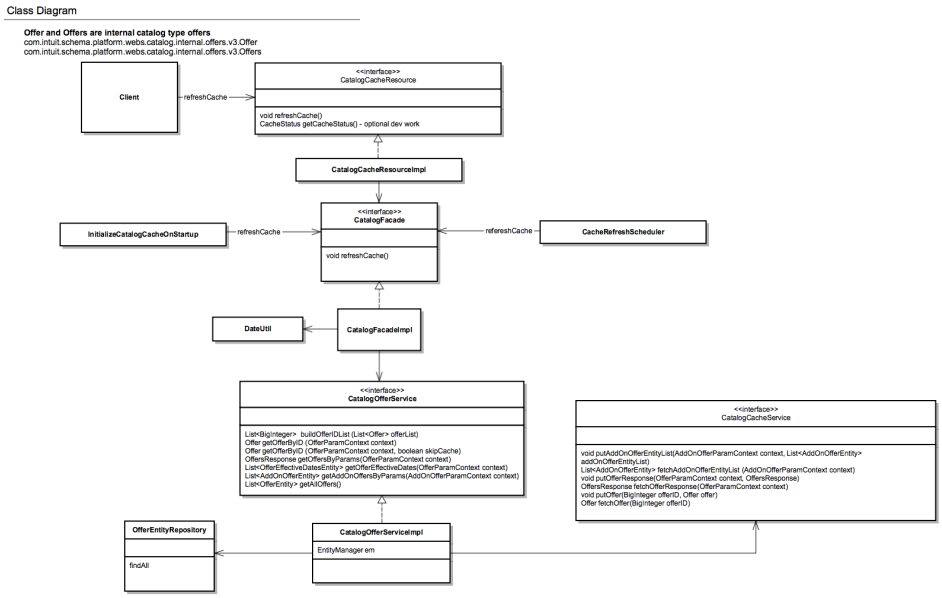
1. Internal Catalog API, on startup preload common addon criteria into CTO couch base cache
2. Internal Catalog API, configured quartz Job, refreshes addon criteria into CTO couch base cache (every 2 hours)
3. Internal Catalog API, configured quartz Job, refreshes query criteria cache into CTO couch base cache (every 12 hours)
4. Internal Catalog API, Public Catalog API, add async end point to refresh the query cache. kickoff job in model library but not wait for end before responding.
5. Call to Get Offers by Query Params Facade
    a. Build Query Param Context
    b. Build Query Using Query Param Context
    c. Verify if query criteria is in CTO couch base cache, if in cache return results
    d. If not in cache, execute query
    e. Build Offers from Query Results
    f. Filter Offers (i.e. Student Offers)
    g. Paginate the list of Offers (limit offers) - Max is 500 *** This is an improvement. Previously, pagination occurred at the end.
    h. Build Offer ID list from list of paginated Offers
    i. In Parallel, call get transitions for Offer IDs (database call) and call get extended attributes for Offer IDs (database call) *** This an improvement. Limit database connections to 1 for transitions. Also, limiting the amount of data returned and processed to no more than 500. Parellization allows for the results to no wait for each other to complete.
    j. Join get transitions results
    k. Join extended attribute results
    l. for each offer
        i. add transitions from transition results
        ii. add extended attributes from extended attribute results
        iii. add addons
        iv. Create addon criteria from offer
        v. Check cache for addon criteria, if it exists add addon results to offer. *** improvement lots of calls to database with same criteria.
        vi. If not exist in cache
            1. Call get addons by addon criteria (database call)
            2. Store criteria in cache, addon critieria as key and addon results
    m. Add query criteria into CTO couch base cache

Class Diagrams

Catalog Model Library



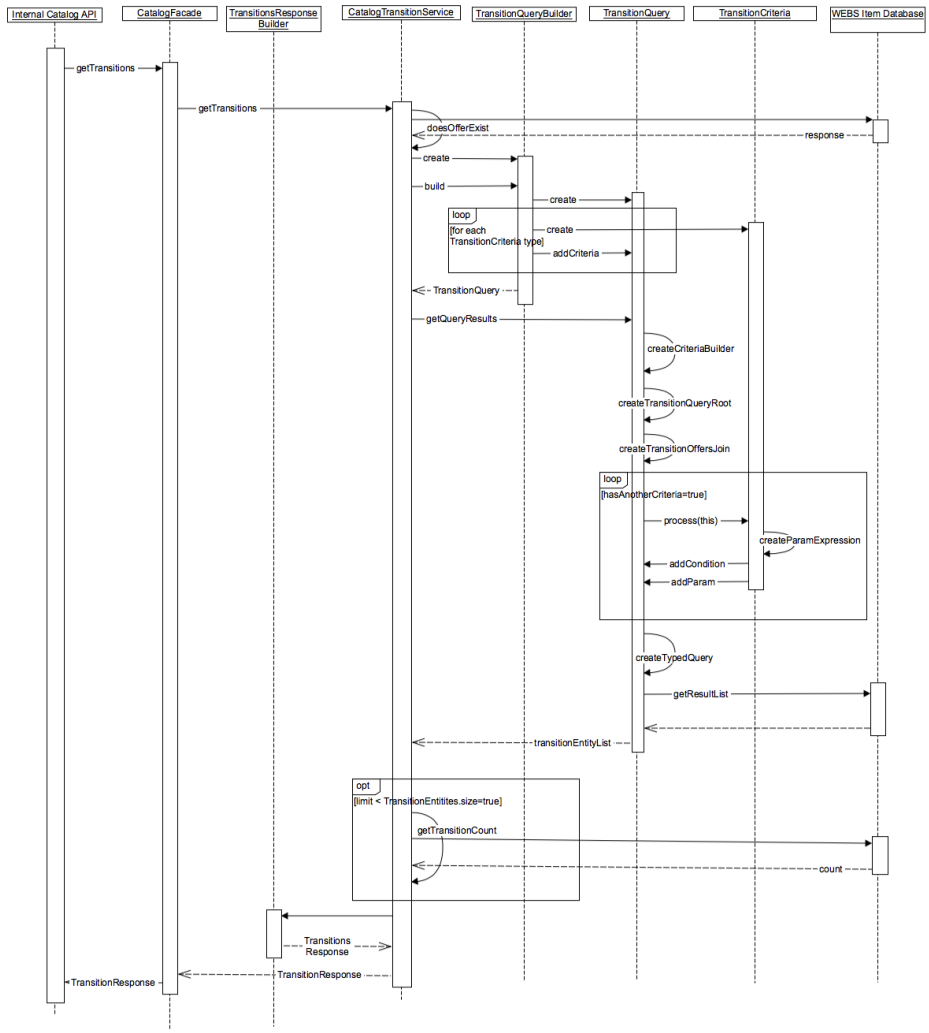
Catalog Cache Manager



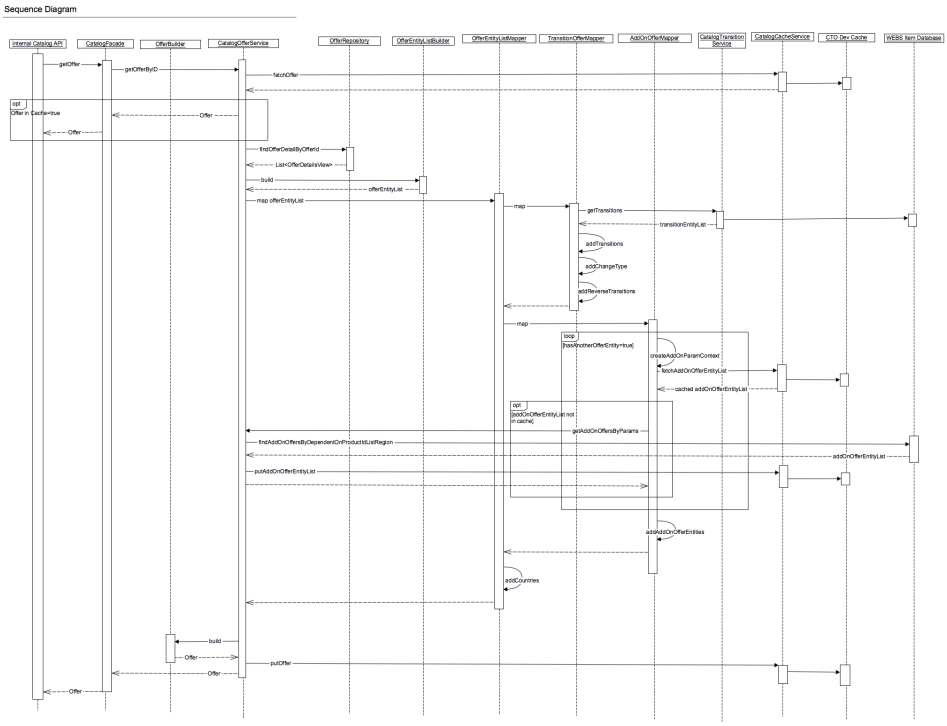
Sequence Diagrams

getTransitions

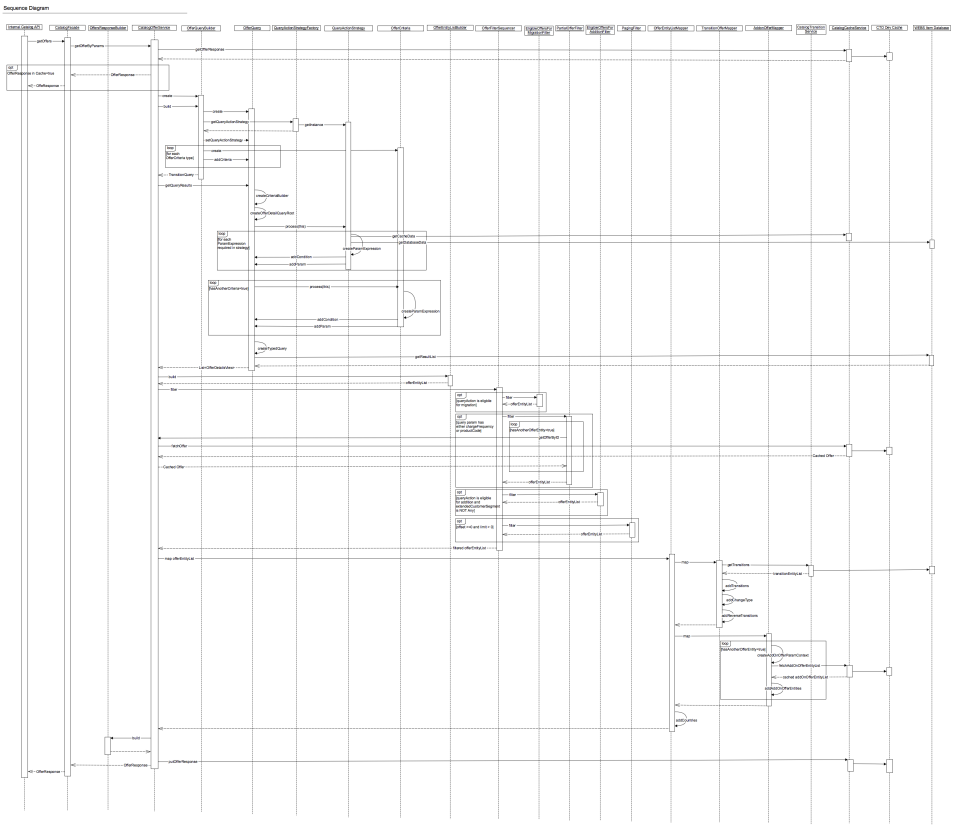
Sequence Diagram



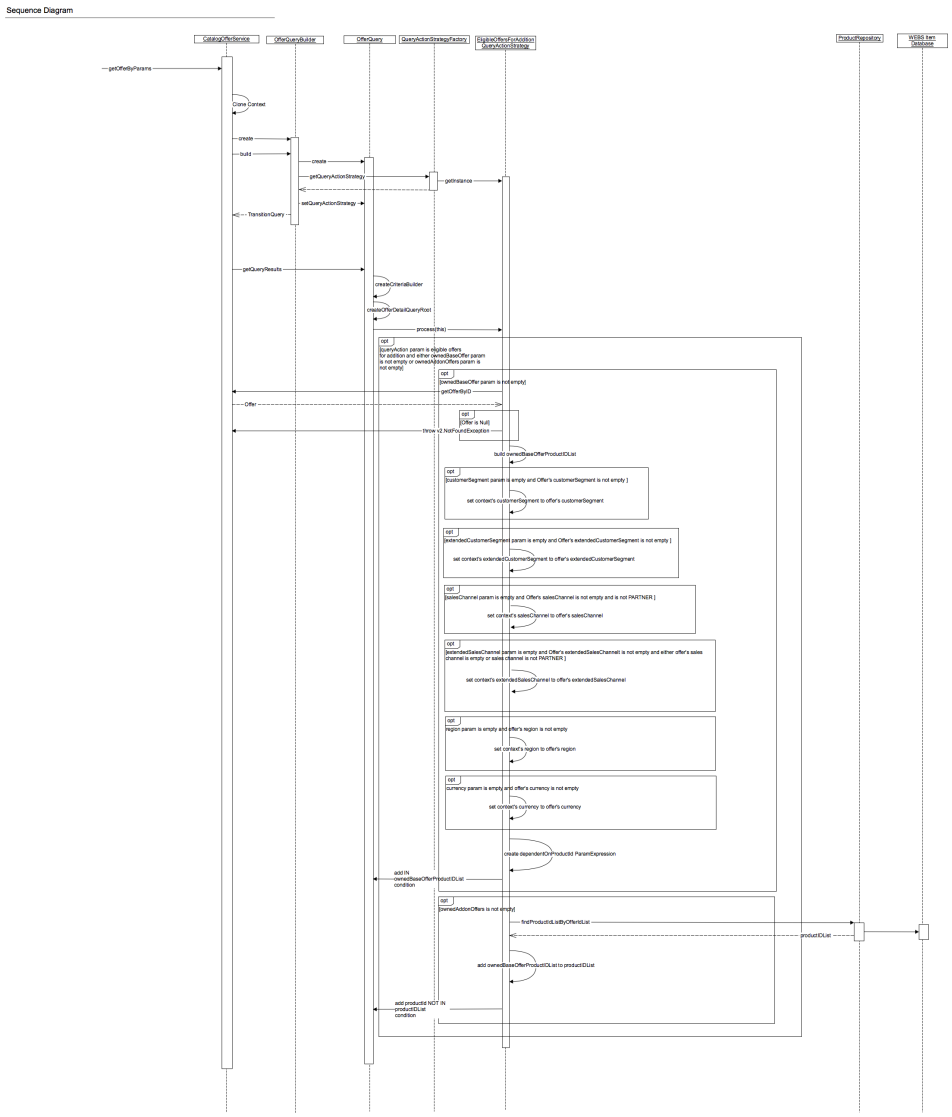
getOffer


Sequence Diagram
Internal Catalog API
CatalogFacade
OfferBuilder
CatalogOfferService
OfferRepositoryImpl
OfferEntityListBuilder
OfferEntityListMapper
TransitionOfferMapper
AddOnOfferMapper
CatalogTransition Service
CatalogCacheService
CTO Dev Cache
WEBS Item Database
getOffer
getOfferByID
fetchOffer
opt
Offer in Cache=true
Offer
findOfferDetailsByOfferId
List<OfferDetailsView>
build
offerEntityList
map offerEntityList
map
getTransitions
transitionEntityList
addTransitions
addChangeType
addReverseTransitions
map
loop
[hasAnotherOfferEntity=true]
createAddOnParamContext
fetchAddOnOfferEntityList
cached addOnOfferEntityList
opt
[addOnOfferEntityList not in cache]
getAddOnOffersByParams
findAddOnOffersByDependentOnProductIdListRegion
addOnOfferEntityList
putAddOnOfferEntityList
addAddOnOfferEntities
addCountries
build
Offer
putOffer
Offer
Offer

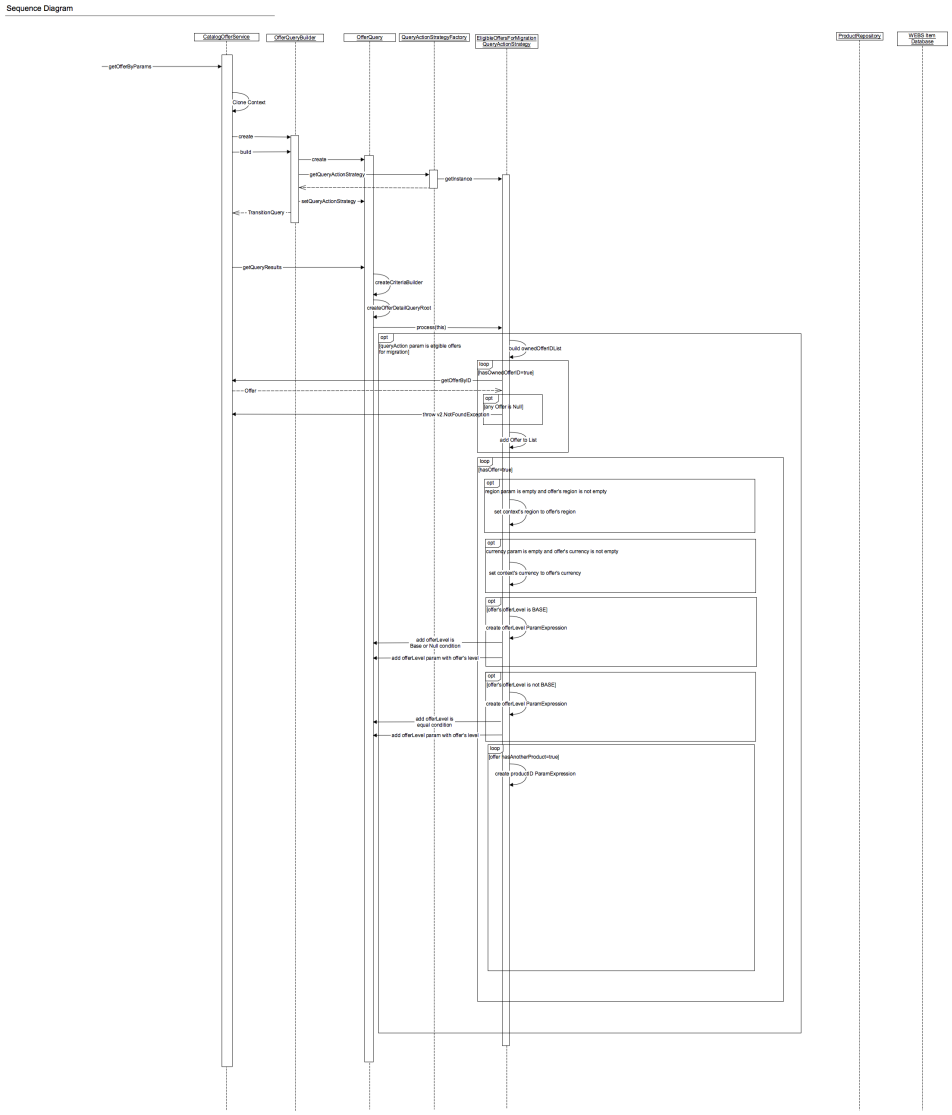
getOffers


Sequence Diagram

GetEligibleOfferForAdditionQueryActionStrategy



GetEligibleOfferForMigrationQueryActionStrategy



GetEligibleOffersFor

GetEligibleOfferForChangeQueryActionStrategy

GetEligibleOfferForSwitchQueryActionStrategy

Load Cache on Private Catalog Cache Manager Startup

Scheduled Private Catalog Cache Refresh

Private Catalog Model Cache Properties

Property	Value	Notes

Profiler Snapshots for Get Offers By Param

Query: offset=0, limit=500, include-inactive-offers=true

Baseline

[snapshot] webs-pri-catalog-offers-profiler.nps

Profiler Snapshot

SQL Query	Total Time		Invocations
select transition0_.TRANSITION_FROM_OFFER_ID as TRANSITION_FROM_OF1_10_, transition0_.TRANSITION_TO_OFFER_ID as TRANSITION_TO_OFFE2_10_, transition0_.TRANSITION_TYPE as TRANSITION_TYPE3_10_, transition0_.DISABLED as DISABLED4_10_ from websitem.offe	53,986 ms	(39.8%)	1,236
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	14,932 ms	(11%)	26
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	9,850 ms	(7.3%)	151
select offerdetai0_.view_rownum as view_rownum1_8_, offerdetai0_.ATTR_NAME as ATTR_NAME2_8_, offerdetai0_.ATTR_TYPE as ATTR_TYPE3_8_, offerdetai0_.BILLING_SERVICE_TYPE as BILLING_SERVICE_TY4_8_, offerdetai0_.CHARGE_DESC as CHARGE_DESC5_8_, offerdetai0_(	6,141 ms	(4.5%)	1
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	4,889 ms	(3.6%)	111
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	4,248 ms	(3.1%)	94
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	4,186 ms	(3.1%)	80
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	2,781 ms	(2.1%)	60
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	2,353 ms	(1.7%)	51
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	2,287 ms	(1.7%)	4
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	2,245 ms	(1.7%)	52
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,980 ms	(1.5%)	44
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,933 ms	(1.4%)	40
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,907 ms	(1.4%)	43
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,895 ms	(1.4%)	43
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,402 ms	(1%)	22
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,398 ms	(1%)	32
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,306 ms	(1%)	27
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	1,082 ms	(0.8%)	25
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	955 ms	(0.7%)	22
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	843 ms	(0.6%)	14
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	831 ms	(0.6%)	20
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	704 ms	(0.5%)	17
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	655 ms	(0.5%)	16
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	645 ms	(0.5%)	15
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	562 ms	(0.4%)	14
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	544 ms	(0.4%)	12
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELECT a.OFFER_ID FROM websitem.offer a, website			
SELECT 1 from DUAL - Database ping	437 ms	(0.3%)	11
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	434 ms	(0.3%)	10
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	424 ms	(0.3%)	10
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	422 ms	(0.3%)	10
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	370 ms	(0.3%)	9
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	352 ms	(0.3%)	8
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	344 ms	(0.3%)	8
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	326 ms	(0.2%)	8
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	316 ms	(0.2%)	8
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	296 ms	(0.2%)	5
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	250 ms	(0.2%)	6
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	243 ms	(0.2%)	6
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	202 ms	(0.1%)	5
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	199 ms	(0.1%)	5
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	182 ms	(0.1%)	4
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	131 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	131 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	126 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	125 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	123 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	122 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	122 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	121 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	121 ms	(0.1%)	3
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	90.2 ms	(0.1%)	2
SELECT * FROM websitem.OFFER_ATTRIBUTE WHERE OFFER_ID in ('30011849', '30011848', '30011851', '30011850', '30011853', '30011852', '30011855', '30011854', '30011841', '30011843', '30011842', '30011845', '30011844', '30011847', '30011846', '30011857', '30011858', '	88.8 ms	(0.1%)	1
SELECT a.OFFER_ID FROM websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e WHERE b.package_id = c.package_id AND a.offer_id = b.offer_id AND b.package_id = c.package_id AND c.product_id = e.product_id AND a.offer_id in (SELEC	82.2 ms	(0.1%)	2

Refactored

Overview | Monitor | Threads | Sampler | Profiler | [snapshot] 11:07:52 AM

Tomcat (pid 68193)

Profiler Snapshot

SQL Query	Total Time		Invocations
select offerdetai0_.view_rownum **as** view_rownum1_12_, offerdetai0_.ATTR_NAME **as** ATTR_NAME2_12_, offerdetai0_.ATTR_TYPE **as** ATTR_TYPE3_12_, offerdetai0_.BILLING_SERVICE_TYPE **as** BILLING_SERVICE_TY4_12_, offerdetai0_.CHARGE_DESC **as** CHARGE_DESC5_12_, of	4,308 ms	(59.6%)	1
select 1 **from** dual - **Database ping**	573 ms	(7.9%)	11
SELECT * **FROM** websitem.offer_transition **WHERE** TRANSITION_FROM_OFFER_ID in ('20000000', '20000001', '20000002', '20000003', '20000004', '20000005', '20000006', '20000007', '20000008', '20000009', '20000011', '20000012', '20000013', '20000014', '20000015', '20	103 ms	(1.4%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	96.8 ms	(1.3%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	88.9 ms	(1.2%)	1
select localizedt0_.ENTITY_ID **as** ENTITY_ID1_7_, localizedt0_.ALT_DESCRIPTION **as** ALT_DESCRIPTION2_7_, localizedt0_.ALT_NAME **as** ALT_NAME3_7_, localizedt0_.COUNTRY_ISO_CODE **as** COUNTRY_ISO_CODE4_7_, localizedt0_.ENTITY_TYPE_CODE **as** ENTITY_TYPE_CODE	79.3 ms	(1.1%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	73.6 ms	(1%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	70.6 ms	(1%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	67.9 ms	(0.9%)	1
select regioncoun0_.COUNTRY_ISO_CODE **as** COUNTRY_ISO_CODE1_11_, regioncoun0_.REGION **as** REGION2_11_ **from** websitem.region_country regioncoun0_ **where** regioncoun0_.REGION='HK'	60.1 ms	(0.8%)	1
SELECT * **FROM** websitem.OFFER_ATTRIBUTE **WHERE** OFFER_ID in ('20000000', '20000001', '20000002', '20000003', '20000004', '20000005', '20000006', '20000007', '20000008', '20000009', '20000011', '20000012', '20000013', '20000014', '20000015', '20000016', '20000	57.6 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	57.5 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	57.2 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	56.6 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	56.4 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	56.4 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	56.4 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	55.8 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	55.6 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	55.6 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	55.5 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	55.1 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (**SELECT** a.OFFER_ID **FROM** websitem.offer a, wel			
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	54.3 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	54.2 ms	(0.8%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	53.8 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	53.7 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	53.4 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	52.9 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	52.6 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	52.0 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	51.2 ms	(0.7%)	1
select regioncoun0_.COUNTRY_ISO_CODE **as** COUNTRY_ISO_CODE1_11_, regioncoun0_.REGION **as** REGION2_11_ **from** websitem.region_country regioncoun0_ **where** regioncoun0_.REGION='ROW'	51.0 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	50.5 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	49.9 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	49.5 ms	(0.7%)	1
select regioncoun0_.COUNTRY_ISO_CODE **as** COUNTRY_ISO_CODE1_11_, regioncoun0_.REGION **as** REGION2_11_ **from** websitem.region_country regioncoun0_ **where** regioncoun0_.REGION='BR'	49.2 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	49.2 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	49.1 ms	(0.7%)	1
SELECT a.OFFER_ID **FROM** websitem.offer a, websitem.offer_component b, websitem.package c, websitem.product e **WHERE** b.package_id = c.package_id **AND** a.offer_id = b.offer_id **AND** b.package_id = c.package_id **AND** c.product_id = e.product_id **AND** a.offer_id in (S	49.0 ms	(0.7%)	1
select regioncoun0_.COUNTRY_ISO_CODE **as** COUNTRY_ISO_CODE1_11_, regioncoun0_.REGION **as** REGION2_11_ **from** websitem.region_country regioncoun0_ **where** regioncoun0_.REGION='CA'	47.7 ms	(0.7%)	1
select regioncoun0_.COUNTRY_ISO_CODE **as** COUNTRY_ISO_CODE1_11_, regioncoun0_.REGION **as** REGION2_11_ **from** websitem.region_country regioncoun0_ **where** regioncoun0_.REGION='GB'	47.6 ms	(0.7%)	1